

§4.1

Basic Concepts of Probability

A number for how likely

Today's Goal

- Define probability and understand what it represents
- Distinguish between theoretical, empirical, and subjective probability
- Apply the classical (theoretical) probability formula
- Apply the relative frequency (empirical) probability formula
- Understand the Law of Large Numbers

What You Will Be Graded On

LT.7 Probability

I understand what the probability of an event represents and can apply basic rules of probability to calculate the probability of an event.

The Forecast Said 70%

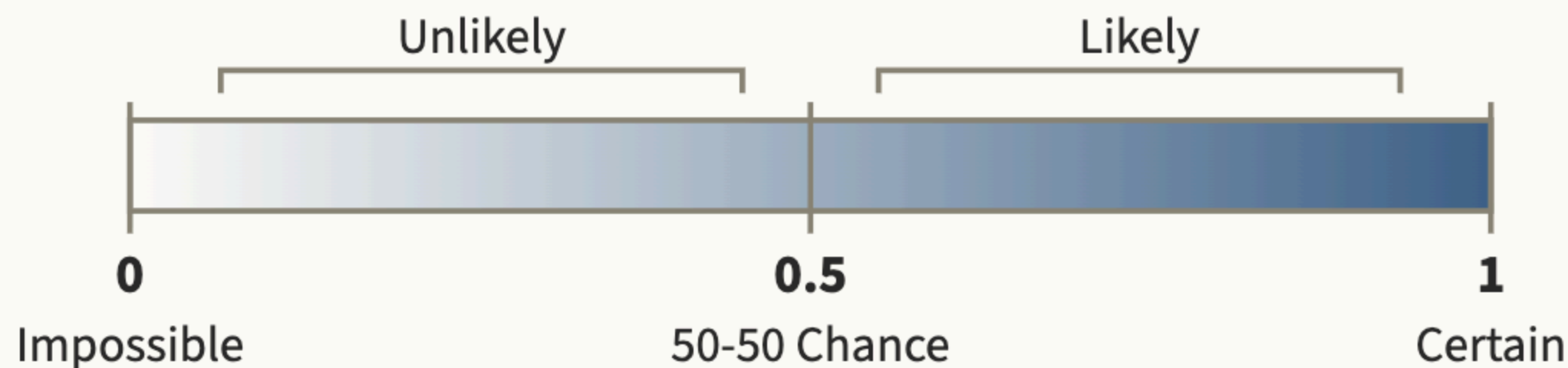
The forecast gives a 70% chance of rain tomorrow. Tomorrow comes, and it stays dry all day.

Was the forecast wrong?

A Number From 0 to 1

Definition 4.1.1

The *probability* of an event is a number between 0 and 1 (inclusive) that represents how likely it is that the event will occur.



Probabilities can be expressed as fractions, decimals, or percentages.

Interpret Each Probability Statement

(a) The probability that a flight arrives on time is 0.85.

There is an 85% chance the flight arrives on time. An on-time arrival is likely but not certain.

(b) The probability of drawing a green card from a standard deck is 0.

It is impossible to draw a green card from a standard deck (which only has red and black cards).

(c) The probability of rolling a number less than 7 on a standard 6-sided die is 1.

When rolling a standard 6-sided die, you are certain to get a number less than 7.

Three Approaches: The First Two

There are three main approaches to determining probability.

Definition 4.1.4

Theoretical probability (also called *classical probability*) is determined by analyzing the possible outcomes of a probability experiment mathematically, assuming all outcomes are equally likely.

Definition 4.1.5

Empirical probability (also called *relative frequency probability*) is determined by conducting an experiment or observing data and calculating the proportion of times an event occurs.

The Third Approach

Definition 4.1.6

Subjective probability is an estimate of probability based on personal judgment, experience, or intuition rather than mathematical analysis or data.

Which type was the 70% rain forecast?

Which Type Is It?

Identify the type of probability being used in each scenario:

(a) A sports analyst gives a team a 60% chance of winning tonight based on their analysis of the matchup and past experience.

Subjective probability. The estimate is based on the analyst's judgment.

(b) A pharmacy finds that 18 out of 600 prescriptions filled needed a correction, so they estimate the correction rate is 3%.

Empirical probability. The probability is based on observed data.

One More Scenario

(c) A student calculates that the probability of drawing an ace from a standard deck of 52 cards is $\frac{4}{52} \approx 0.0769$.

Theoretical probability. The probability is calculated mathematically based on the structure of a deck of cards.

What one question tells the three types apart?

Where Does the Number Come From?

- From the structure of the experiment: **theoretical**.
- From observed data: **empirical**.
- From personal judgment: **subjective**.

Experiments and Outcomes

To calculate theoretical probability, we need to understand some basic terminology.

Definition 4.1.8

A *probability experiment* (or *random experiment*) is a procedure that can be repeated and has a well-defined set of possible outcomes.

Definition 4.1.9

An *outcome* is a single result of a probability experiment.

Sample Spaces and Events

Definition 4.1.10

The *sample space* of a probability experiment, denoted by S , is the set of all possible outcomes.

Definition 4.1.11

An *event* is a subset of the sample space—that is, a collection of one or more outcomes.

Writing Out a Sample Space

Identify the sample space for each probability experiment:

- (a) Spinning a spinner with 5 equal sections numbered 1 through 5.

$$S = \{1, 2, 3, 4, 5\}.$$

- (b) Selecting a card from a standard deck and recording its color.

$$S = \{\text{Red}, \text{Black}\}.$$

- (c) Answering two true/false questions.

$$S = \{(T, T), (T, F), (F, T), (F, F)\} \text{ where } T = \text{true and } F = \text{false}.$$

Count the Outcomes in the Event

The formula for theoretical probability assumes that all outcomes in the sample space are equally likely:

$$P(E) = \frac{\text{Number of outcomes in event } E}{\text{Total number of outcomes in the sample space}}.$$

When would this formula give the wrong answer?

When the outcomes are not equally likely, such as a loaded die or a biased coin.

One Die, Three Events

A fair 12-sided die, numbered 1 through 12, is rolled. Find the probability of:

(a) Rolling a 9.

The sample space is $S = \{1, 2, 3, \dots, 12\}$, with 12 equally likely outcomes. There is 1 favorable outcome:

$$P(\text{rolling a 9}) = \frac{1}{12} \approx 0.083 \text{ or } 8.3\%.$$

Events With Several Outcomes

The same 12-sided die is rolled. Find the probability of:

(b) Rolling a multiple of 3.

The event is $\{3, 6, 9, 12\}$, with 4 favorable outcomes:

$$P(\text{multiple of 3}) = \frac{4}{12} = \frac{1}{3} \approx 0.333 \text{ or } 33.3\%.$$

(c) Rolling a number greater than 9.

The event is $\{10, 11, 12\}$, with 3 favorable outcomes:

$$P(\text{greater than 9}) = \frac{3}{12} = \frac{1}{4} = 0.25 \text{ or } 25\%.$$

Now You Try: A Deck of Cards

You Try 4.1.1. A card is randomly selected from a standard deck of 52 cards. Find the probability of each event. Round decimal answers to 4 decimal places.

- (a) Selecting a king
- (b) Selecting a red card
- (c) Selecting a face card (jack, queen, or king)

A standard deck: 4 suits of 13 cards. Hearts and diamonds are red, clubs and spades are black.

You Try 4.1.1: Checking Your Answers

(a) Selecting a king

$$P(\text{king}) = \frac{4}{52} = \frac{1}{13} \approx 0.0769$$

(b) Selecting a red card

$$P(\text{red}) = \frac{26}{52} = \frac{1}{2} = 0.5$$

(c) Selecting a face card (jack, queen, or king)

$$P(\text{face card}) = \frac{12}{52} = \frac{3}{13} \approx 0.2308$$

Probability From Observed Data

When outcomes are not equally likely, or when we want to estimate probability from observed data, we use empirical probability:

$$P(E) = \frac{\text{Number of times event } E \text{ occurred}}{\text{Total number of trials or observations}}.$$

A delivery driver arrived on time for 171 of 190 deliveries last month. What is the probability that the next delivery arrives on time?

$P(\text{arrives on time}) = \frac{171}{190} = 0.9$ or 90%. Based on past performance, there is a 90% chance the next delivery will arrive on time.

Ten Heads in a Row

A fair coin lands on heads 10 times in a row.

**What do you bet on
for the next flip?**

The Law of Large Numbers

Theorem 4.1.15 Law of Large Numbers

As the number of trials of a probability experiment increases, the empirical probability of an event approaches the theoretical probability of the event.

Flip a fair coin just 20 times, and we might get 13 heads (empirical probability $= 0.65$), which seems quite different from 0.5.

Flip the coin 2,000 times, and we would expect the proportion of heads to be much closer to 0.5.

Why are casinos profitable in the long run?

The Coin Has No Memory

Warning 4.1.17

The Law of Large Numbers does *not* mean that short-term deviations will “balance out” in the near future. This misconception is known as the *gambler’s fallacy*. For example, if a fair coin lands on heads 10 times in a row, the probability of the next flip being tails is still 50%—the coin has no “memory” of previous flips.

Three Properties and a Cutoff

We use $P(E)$ to denote the probability of event E . All probabilities must satisfy the following properties:

1. $0 \leq P(E) \leq 1$ for any event E .
2. $P(S) = 1$ where S is the sample space (something must happen).
3. $P(\emptyset) = 0$ where $\emptyset$ is the empty set (an impossible event).

Definition 4.1.18

An event is considered *unusual* if its probability is very low—typically less than or equal to 0.05 (5%).

Unusual or Not?

Determine whether each event is unusual:

(a) A randomly chosen person shares your birthday. Assume all 365 birthdays are equally likely.

$P(\text{shares your birthday}) = \frac{1}{365} \approx 0.0027 < 0.05$. **This is an unusual event.**

(b) Winning a prize in a raffle with 250 tickets sold, where you purchased 20 tickets.

$P(\text{winning}) = \frac{20}{250} = 0.08 > 0.05$. **This is not an unusual event.**

What to Take With You

- A probability is a number from 0 to 1. It says how likely, not what will happen.
- Theoretical: count equally likely outcomes. Empirical: use observed data.
Subjective: use judgment.
- More trials bring the empirical probability closer to the theoretical one. The coin still has no memory.
- An event with probability 0.05 or less is unusual.